

Project 3 Longitudinal Data Analysis

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Abstract

Longitudinal studies investigating the progression of Alzheimer’s Disease (AD) are frequently compromised by non-random participant attrition, leading to missing data patterns that challenge the validity of clinical inferences. This study analyzes the disease trajectory of 1,253 patients followed over a seven-year period, focusing on cognitive decline as measured by the Clinical Dementia Rating Sum of Boxes (CDRSB), psychiatric symptoms via the Brief Psychiatric Rating Scale (BPRS), and associated biomarker profiles. To address the limitations of standard complete-case analyses, we implemented a comprehensive methodological comparison across three modeling frameworks: Linear Mixed-Effects (LME) models for continuous outcomes, and Generalized Estimating Equations (GEE) alongside Generalized Linear Mixed Models (GLMM) for categorical indicators.

Our results highlight the importance of appropriately handling missing data when modeling disease progression. While differences across modeling strategies are not always sharply delineated, the analyses indicate that GEE is valid only under a Missing Completely At Random (MCAR) assumption and do not properly account for Missing At Random (MAR) data; therefore, Multiple Imputation (MI) is required to obtain reliable inference under MAR. In contrast, LME and GLMM are generally robust under a MAR assumption and can handle such missingness more effectively. Moreover, a sensitivity analysis suggests that a Missing Not At Random (MNAR) mechanism does not appear to affect significantly our data. Thus, the use of methods capable of directly handling at least MAR missingness, together with appropriate sensitivity analyses, are essential in AD research to avoid potentially misleading conclusions.

Contents

List of Figures	3
List of Tables	3
List of Abbreviations	4
1 Introduction	5
2 Preliminary Data Exploration	5
2.1 Exploratory Analysis of Missing Data Patterns	6
2.2 Statistical Assessment of the Dropout Mechanism	9
3 Assessment of Missing Data Mechanisms	11
3.1 Impact of the MAR Assumption	11
3.1.1 Linear Mixed Model	11
3.1.2 Marginal Model	13
3.1.3 Random-Effects Model	16
3.2 Sensitivity Analysis regarding MNAR in the LME model	21
4 Conclusion and further studies	22
5 References	24
A Appendix	26

List of Figures

1	Percentage of Missing Values in the CDRSB measurement.	7
2	BPRS' mean value's evolution with respect to dropout patterns.	7
3	Dichotomized CDRSB's evolution with respect to dropout patterns.	8
4	BPRS: Observed vs Missing at next visit.	8
5	Dichotomized CDRSB: Observed vs Missing at next visit.	9
6	Distribution of Missingness by ADL and WZC.	10
7	Empirical and model-based variance patterns over time (LME).	12
8	Dichotomized CDRSB's Empirical Variance vs. Fitted Variance (MI-GLMM).	19
9	Observed probability trajectory vs. fitted model prediction (MI-GLMM).	20
10	Missingness by Baseline Covariates.	27

List of Tables

1	Logistic Regression Analysis of Dropout Predictors.	10
2	LME vs. MI-LME ($M = 10$).	13
3	GEE vs. MI-GEE ($M = 20$).	16
4	MI-GEE model ($M = 20$) p -value < 0.1	17
5	GLMM vs. MI-GLMM ($M = 10$).	18
6	SA: Estimates of relevant parameters.	22
7	MI-GEE model ($M = 20$).	26
8	MI-GEE Model ($M = 20$) on the reduced model.	28
9	SA: Estimates of parameters.	29

List of Abbreviations

ACMV	Available Case Missing Values
AD	Alzheimer's Disease
CCMV	Complete Case Missing Values
CI	Confidence Interval
GEE	Generalized Estimating Equations
GLMM	Generalized Linear Mixed-Effects Model
LME	Linear Mixed-Effects
LRT	Likelihood Ratio Test
MAR	Missing At Random
MCAR	Missing Completely At Random
MI	Multiple Imputation
MI-GEE	Generalized Estimating Equations with Multiple Imputation
MI-GLMM	Generalized Linear Mixed-Effects Model with Multiple Imputation
MI-LME	Linear Mixed-Effects with Multiple Imputation
MNAR	Missing Not At Random
NCMV	Neighbouring Case Missing Values
OLS	Ordinary Least Squares
PMM	Pattern-Mixture Model
QIC	Quasi-Likelihood Information Criterion
SA	Sensitivity Analysis
SE	Standard Error
STD	Standard Delta-Adjustment
WGEE	Weighted Generalized Estimating Equations

1 Introduction

Alzheimer’s disease (AD) is one of the most prevalent forms of late-life dementia, representing a major public health challenge with profound social and economic consequences [2]. Characterized by progressive cognitive decline and psychiatric symptoms, AD significantly impacts patients’ daily functioning and quality of life, while also placing a considerable burden on caregivers and healthcare systems.

The aim of this report is to account for missing data in a more rigorous way and to investigate whether the substantive conclusions change compared to analyses that ignore missingness. In earlier analyses, models were fitted using only the observed data, implicitly assuming that missing values do not affect the results. In this report, missingness is handled explicitly using methods that are valid under the Missing At Random (MAR) assumption, in particular Multiple Imputation (MI) and models that appropriately account for incomplete longitudinal data. In addition, a sensitivity analysis is performed for the continuous outcome to assess how robust the main conclusions are to potential violations of the MAR assumption, allowing for the possibility of Missing Not At Random (MNAR) mechanisms. By comparing results obtained under different assumptions about the missing data process, we evaluate whether ignoring missingness or misspecifying the missingness mechanism may affect parameter estimates, uncertainty, or overall conclusions. This approach provides a more complete assessment of the impact of missing data on the interpretation of the Alzheimer study results.

The code used to get our results can be retrieved in [5].

2 Preliminary Data Exploration

The empirical foundation of this study is a comprehensive longitudinal dataset comprising 1,253 unique patients followed up over a seven-year period (baseline plus six annual visits). The data structure is multidimensional, capturing static baseline demographic and clinical characteristics (e.g., **age**, **sex**, **edu**, **bmi**...) alongside dynamic, time-varying measurements for primary outcomes. These longitudinal indicators include the Brief Psychiatric Rating Scale (**bprs**), the Clinical Dementia Rating Sum of Boxes (**cdrs**), and biomarker profiles derived from PET scans for Amyloid-Beta (**abpet**) and Tau protein (**taupet**):

patid: A unique identifier for each patient.

trial: A categorical variable (1-25) indicating the clinical trial or center.

sex: Categorical (0 = Male, 1 = Female).

age: Patient’s age at baseline.

edu: Categorical education level (1 = Primary, 2 = Lower Secondary, 3 = Upper Secondary, 4 = Higher).

bmi: Body Mass Index at baseline.

inkomen: Income at baseline.

job: Categorical (0 = No Job, 1 = Job).

adl: Activities of Daily Living score at baseline.

wzc: Categorical residence (0 = Home, 1 = Residence).

bprs0...bprs6: Brief Psychiatric Rating Scale (annual measurements).

cdrsb0...cdrsb6: Clinical Dementia Rating Scale Sum of Boxes (annual measurements).

CDRSB_cat: Binarized version of CDRSB:

$$\text{CDRSB_cat} = \begin{cases} 0 & \text{se CDRSB} \leq 10, \\ 1 & \text{se CDRSB} > 10. \end{cases}$$

abpet0...abpet6: Amyloid-Beta PET scan results (annual measurements).

taupet0...taupet6: Tau PET scan results (annual measurements).

2.1 Exploratory Analysis of Missing Data Patterns

Given the longitudinal nature of these parameters, we started the analysis by performing an exploratory investigation of the missing data patterns. First of all, is important to note that the dropout pattern is identical for the CDRSB and BPRS scores, indicating that missingness affects both outcome measures in the study in the same way. Therefore, all conclusions drawn from the analysis of missing data can be considered generalizable to both outcome measures. Afterwards, we assessed whether the missingness was monotone. Using a dedicated function in R, we confirmed that the missingness pattern is indeed monotone. This can be seen in Figure 1, where a line for a subject begins at the visit where the observation is first missing and continues for all subsequent missing visits, illustrating the monotone missingness pattern. In other words, once a subject has a missing observation at a given visit, all subsequent measurements for that subject are also missing. Identifying a monotone missingness structure is important because it simplifies the implementation of methods such as MI. Figure 1 is also informative regarding the percentages of dropout per visit. It can be observed that the proportion of missing observations increases progressively with each subsequent visit, reflecting the typical pattern of participant dropout in longitudinal studies. Moreover the percentage of missing observations is 29.1%.

The observed dropout patterns in the BPRS score, which are reported in Figure 2, shows important aspects regarding the missing data mechanism in this study. Patients who drop out earlier of the study exhibit systematically higher symptom levels at baseline and during follow-up than those who remain under observation. In addition, the evolution of the BPRS means appears linear across measurement occasions for each dropout group. This association between dropout time and outcome trajectory is incompatible with a Missing Completely At Random (MCAR) mechanism, under which the probability of missingness should be unrelated to both observed and unobserved outcomes. Consequently, the missingness process is likely to be non-ignorable.

Figure 3 shows the evolution of the dichotomized CDRSB outcome stratified by dropout time. In contrast to the continuous BPRS scores, it does not exhibit a clear linear pattern over time, but it shows the classical S-shape that is expected when modeling probabilities.

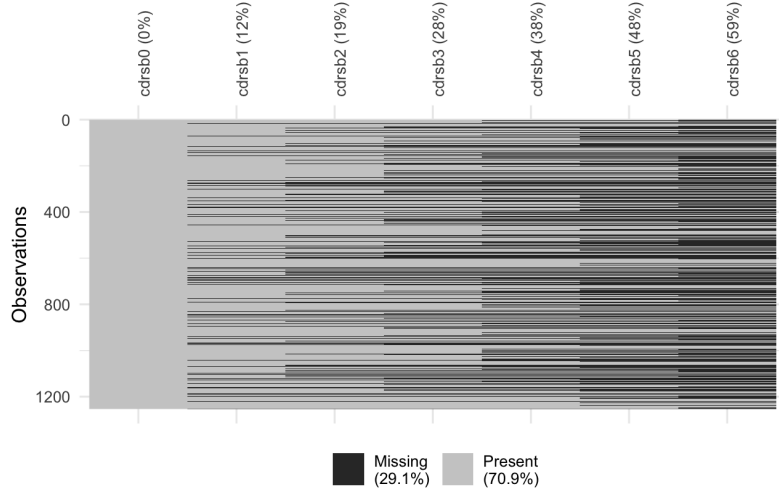


Figure 1: Percentage of Missing Values in the CDRSB measurement.

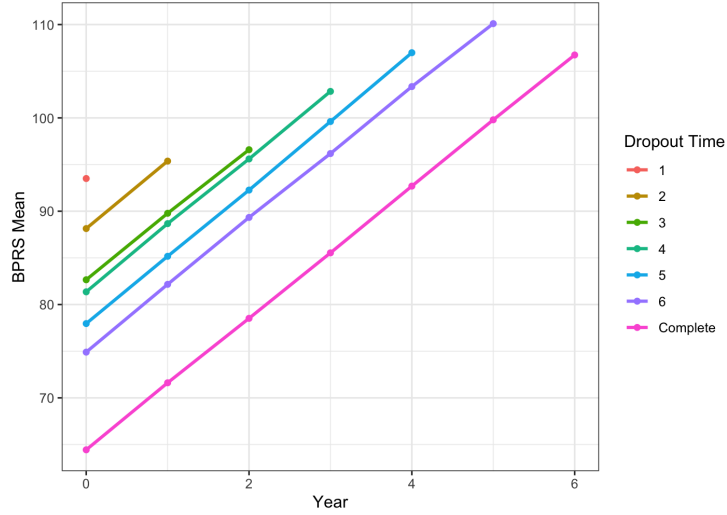


Figure 2: BPRS' mean value's evolution with respect to dropout patterns.

Moreover, it is less straightforward from the data to conclude whether participants who start with less severe symptoms are more likely to reach the final visit (timepoint 6) or not.

At this point of the analysis it is interesting to understand which mechanism of missingness could explain our intuitions. Figure 4 shows for any time t a comparison between the mean of the BPRS score in patients who remain in the study at time $t + 1$ and patients who drop out at time $t + 1$. We noticeably observe a persistent separation between the two groups (red bars, the ones linked to the drop-out patients, systematically exceed the blue ones, linked to patients who stay in the study). This result highly suggests that the missingness pattern depends on observed outcome values: we can conclude that an MCAR mechanism is not appropriate for the BPRS score, while a MAR mechanism, if not MNAR, seems more coherent with these results.

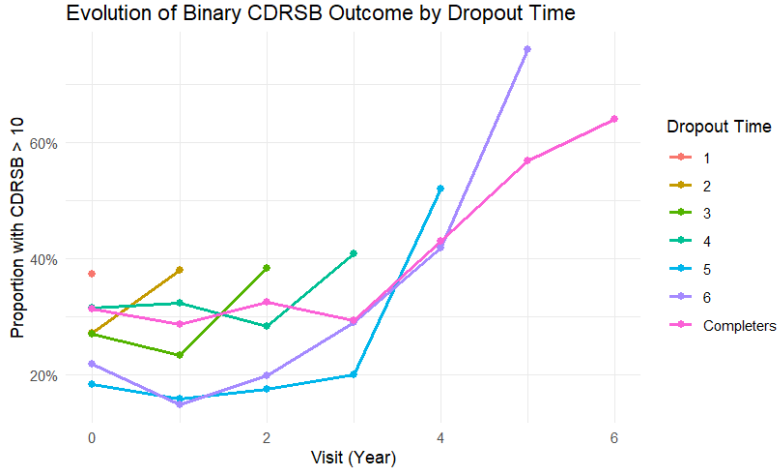


Figure 3: Dichotomized CDRSB's evolution with respect to dropout patterns.

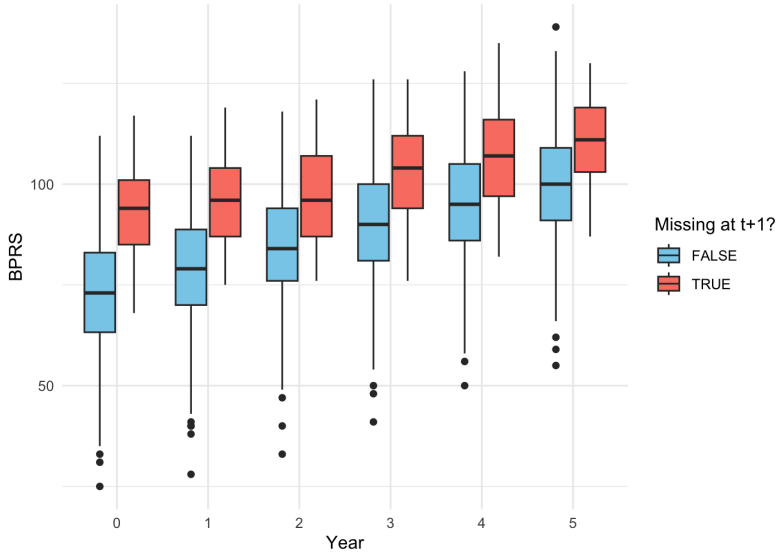


Figure 4: BPRS: Observed vs Missing at next visit.

The same results can be drawn for the CDRSB score. Figure 5 shows for any time t the fraction of patients with a CDRSB value bigger than 10. More in detail, the red line represents the patients that, for any time t , will drop out at time $t + 1$, while the blue line represents the patients that will stay in the study. As it is clearly noticeable, this last fraction is consistently lower than the dropout one. This indicates that participants who are more likely to have missing data at the next visit also have a higher probability of exceeding the CDRSB threshold at the current visit. Therefore, the assumption of MCAR on the dichotomized CDRSB value does not appear to hold true. This suggests that we should take into consideration (at least) a MAR mechanism for this outcome, and a sensitivity analysis should be performed to distinguish whether an MNAR mechanism is more appropriate or not.

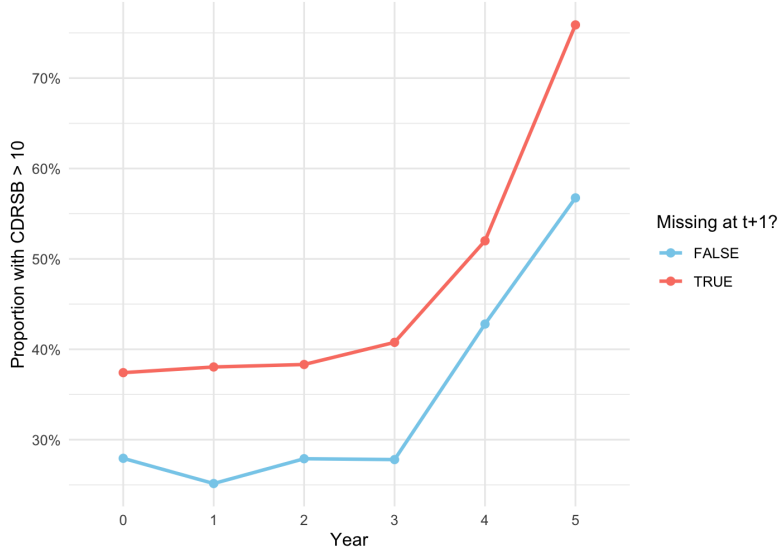


Figure 5: Dichotomized CDRSB: Observed vs Missing at next visit.

Finally, to comprehensively characterize the missing data mechanism we investigated the potential association between dropout and baseline covariates available in the dataset. These results are extremely useful, especially in the light of a MI task. In order to proceed, we visually inspected the missingness rates stratified by various baseline characteristics (see Figure 10 in the Appendix). We report a detailed example of the results for the ADL and WZC variables in Figure 6. Specifically, the left panel shows that participants with lower ADL scores are markedly more likely to drop out earlier in the study compared to those with higher scores. Similarly, the right panel reveals that participants residing in a residential care center (WZC=1) exhibit a higher tendency for early dropout compared to those living at home (WZC=0). In general terms, this exploratory analysis revealed that variables such as AGE, JOB, ADL, and WZC are associated with the probability of attrition. Conversely, other covariates such as TRIAL, EDU, BMI, and INKOMEN did not appear to have a substantial influence on the missingness process. These observed disparities in dropout rates underscore the importance of conditioning on these specific baseline characteristics to mitigate potential selection bias in the analysis.

2.2 Statistical Assessment of the Dropout Mechanism

To corroborate the visual findings with statistical evidence, we fitted a logistic regression model to predict the probability of missingness at a given visit based on previous observed outcomes and baseline covariates. Specifically, we modeled the binary indicator of missingness ($R_{ij} = 1$ if missing, 0 otherwise) as a function of the lagged outcome measures (observed values at the previous visit), time, and baseline characteristics.

Table 1 presents the results of this analysis (only the selected significant parameters are listed). The model confirms that the missingness mechanism is strongly related to the patient’s history. Crucially, the previous measurement of the primary outcome, `prev_cdrsb`, is a highly significant predictor of dropout ($p < 0.001$). The positive coefficient ($\beta = 0.121$) indicates that patients with higher CDRSB scores (indicating worse cognitive function)

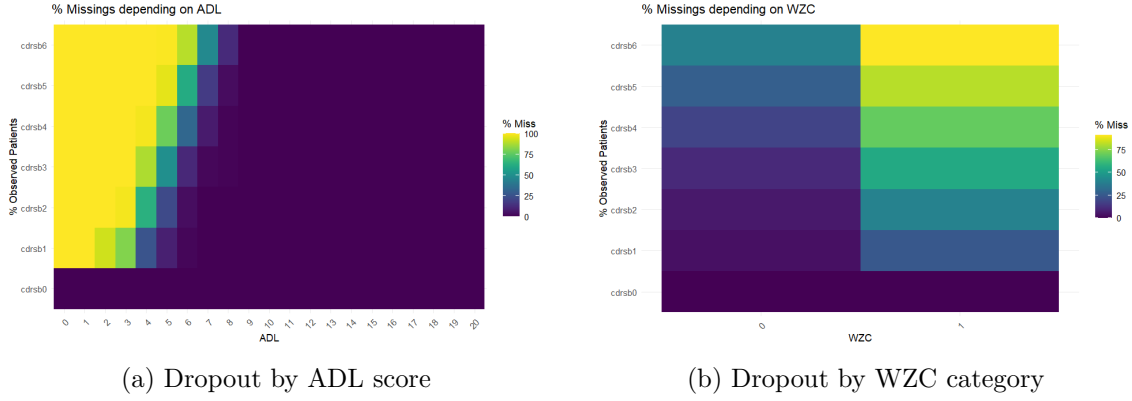


Figure 6: Distribution of Missingness by Baseline Covariates for ADL and WZC.

at the previous visit are significantly more likely to have missing data at the subsequent visit.

Furthermore, the model reinforces the importance of the baseline covariates identified in the exploratory plots. Residence in a care center (WZC) is strongly associated with increased odds of dropout ($\beta = 1.984, p < 0.001$), while higher functional independence scores (ADL) are associated with a lower probability of missingness ($\beta = -1.971, p < 0.001$).

Table 1: Logistic Regression Analysis of Dropout Predictors.

Parameter	Estimate	Std. Error	Pr(> z)
prev_cdrsb	0.121	0.011	< 0.001
ADL	-1.971	0.191	< 0.001
WZC	1.984	0.144	< 0.001
TIME	1.441	0.142	< 0.001

To conclude, the statistical significance of the previous outcome (**prev_cdrsb**) provides strong evidence against the MCAR assumption. Since the probability of missingness depends on observed information (specifically, the previous health status of the patient), the mechanism satisfies the conditions for MAR. Consequently, analyses that rely on the MAR assumption, such as MI, are appropriate and necessary for this dataset to minimize bias.

3 Assessment of Missing Data Mechanisms

3.1 Impact of the MAR Assumption

In this section we want to study how an hypothesis of MAR on our data would impact the results that we got in the previous analyses ([6] and [7]).

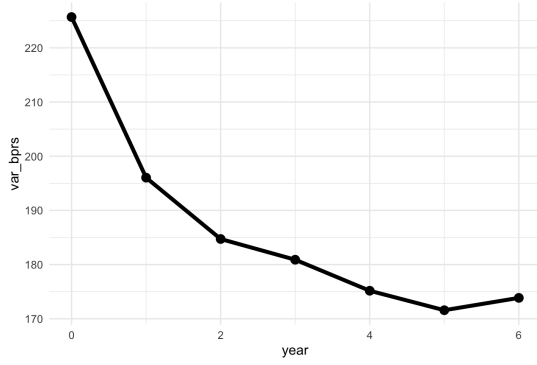
3.1.1 Linear Mixed Model

To account for the MAR assumption, a Linear Mixed-Effects (LME) model with BPRS score as the outcome variable was adopted. According to the literature, under the MAR assumption, parameter estimates obtained from mixed-effects models are consistent and unbiased [12]. In our previous analysis [6], this model had already been fitted. We first assessed the necessity of including a random intercept and a random slope for each patient. Subsequently, starting from a full model, a stepwise model selection procedure based on Likelihood Ratio Test (LRT) was applied, leading to the final model.

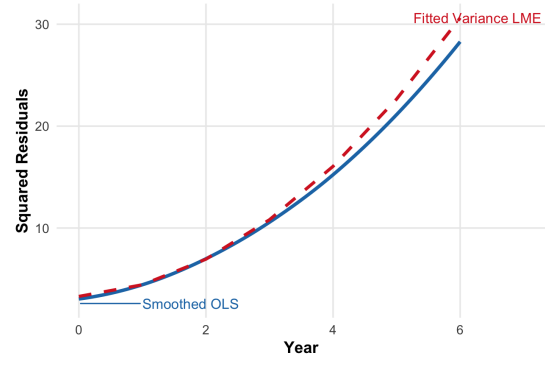
$$\left\{ \begin{array}{l} \text{BPRS}_i = \beta_0 + \beta_{0,T}\text{Trial}_i + \beta_{0,S}\text{Sex}_i + \beta_{0,A}\text{Age}_i + \beta_{0,E}\text{Edu}_i + \beta_{0,B}\text{Bmi}_i \\ \quad + \beta_{0,J}\text{Job}_i + \beta_{0,AD}\text{Adl}_i + \beta_{0,W}\text{WZC}_i + \beta_{0,C}\text{CDRSB}_i \\ \quad + (\beta_1 + \beta_{1,A}\text{Age}_i + \beta_{1,E}\text{Edu}_i + \beta_{1,J}\text{Job}_i + \beta_{1,AD}\text{Adl}_i + \beta_{1,C}\text{CDRSB}_i) \cdot \text{Year}_i \\ \quad + b_{0i} + b_{1i}\text{Year}_i + \varepsilon_i \\ b_i \sim \mathcal{N}(0, D) \\ \varepsilon_i \sim \mathcal{N}(0, \Sigma) \\ \Sigma = \sigma^2 \cdot I_{n_i} \end{array} \right. \quad (1)$$

It is important to note that the empirical variance of the BPRS variable, computed directly at each time point (Figure 7a), represents the total observed variability in the data, without conditioning on any covariates or assuming a specific model structure. In contrast, the variance implied by the LME model arises from a model-based decomposition of the marginal variance, distinguishing the between-subject variability captured by the random effects from the within-subject residual variability after accounting for the fixed effects. As shown in Figure 7, although the total empirical variance tends to decrease over time, the variance predicted by the LME model exhibits a slightly increasing trend. This apparent discrepancy does not indicate model misspecification, as a more meaningful comparison involves the LME-predicted within-subject variance and the smoothed squared residuals from an Ordinary Least Squares (OLS) model. The close agreement between these two curves indicates the appropriateness of the selected random effects.

After that, although the linear mixed-effects model is likelihood-based and therefore theoretically sufficient to handle MAR data, we still applied MI to further strengthen the analysis. As discussed by Verbeke and Molenberghs [12], when several plausible mechanisms for missingness can be postulated and one wishes to account for all of them, MI can be used as a solution to handle each of these assumptions. In our context, while MAR remains the primary working assumption, the longitudinal structure of the data and the occurrence of informative dropout suggest a possible MNAR mechanism, which we can handle by performing MI to account for potential deviations from MAR. To ensure a fair comparison between models with and without MI, we began with a full model for both



(a) Variance of BPRS over time



(b) Smoothed average of squared OLS residuals vs Fitted LME variance function

Figure 7: Empirical and model-based variance patterns over time (LME).

analyses. Starting from a reduced model would have been problematic because applying variable selection separately to the MI and non-MI datasets could have produced two different reduced models, making them non-comparable. By beginning with identical full models, we guaranteed that any differences observed were attributable solely to the use of MI. With regard to the number of imputations to be carried out, well-established results reported by Molenberghs and Verbeke [12] indicate that a relatively limited number of imputations, such as 5 or 10, is generally adequate to obtain reliable results. Results from more recent studies (such as Graham et al., 2007 [8]) suggest to use higher values, but due to convergence problems 10 multiple imputations were adopted in this part of the analysis.

As it can be observed from the estimates, their corresponding standard errors, and the associated p -values reported in Table 2, the differences between the coefficients obtained from the standard LME model and those obtained from the Linear Mixed-Effects with Multiple Imputation (MI-LME) model are minimal and not practically meaningful. This suggests that, in line with theoretical expectations, the LME model without MI remains robust to missing data, providing reliable parameter estimates even in the presence of incomplete observations. Moreover, the close agreement in standard errors and p -values across the two approaches indicates that the MI-LME model yields stable and reliable inference while appropriately accounting for missing data. Therefore, the application of MI in this context does not substantially alter the results, supporting the robustness of the standard LME approach under the observed missingness pattern. This view is further supported by Huque et al. [9], who point out that MI is not required when covariates are fully observed, as is the case in our study. In particular, key covariates such as **TIME**, **AGE**, **BMI**, **WZC**, and **CDRSB0** show similar effects across models and remain statistically significant, as do several time-covariate interactions (e.g. **TIME*ADL**, and **TIME*CDRSB0**). Nevertheless, standard errors and p -values are generally comparable, suggesting that the MI-LME model provides stable and reliable inference while appropriately accounting for missing data.

Table 2: Comparison of the results for the MI-LME model (with $M = 10$ imputations) and LME model, excluding TRIAL effects.

Parameter	MI-LME ($M = 10$)		LME	
	Estimate (SE)	p -value	Estimate (SE)	p -value
Intercept	-53.977 (4.831)	< .0001	-47.733 (4.426)	< .0001
TIME	6.618 (1.701)	0.0018	5.216 (2.235)	0.0198
SEX	-0.140 (0.148)	0.3449	-0.170 (0.137)	0.2158
AGE	1.708 (0.046)	< .0001	1.636 (0.043)	< .0001
EDU 2	0.149 (0.205)	0.4712	0.115 (0.171)	0.502
EDU 3	-0.089 (0.239)	0.7096	0.006 (0.216)	0.979
EDU 4	0.078 (0.319)	0.8079	0.129 (0.288)	0.655
BMI	0.194 (0.029)	< .0001	0.157 (0.029)	< .0001
JOB 1	-5.286 (0.894)	< .0001	-3.997 (0.834)	< .0001
ADL	0.089 (0.132)	0.5029	-0.105 (0.122)	0.3900
WZC 1	2.037 (0.146)	< .0001	1.922 (0.100)	< .0001
CDRSB0	-0.017 (0.006)	0.0050	-0.020 (0.004)	< .0001
ABPET0	0.039 (0.223)	0.8610	0.037 (0.182)	0.8392
TAUPET0	-0.281 (0.602)	0.6439	0.328 (0.484)	0.4986
INKOMEN	0.000036 (0.000217)	0.8674	-0.00000374 (0.000207)	0.9856
TIME*INKOMEN	0.000044 (0.000103)	0.6739	0.000095 (0.000093)	0.3077
TIME*ABPET0	-0.112 (0.138)	0.4213	-0.089 (0.121)	0.4643
TIME*TAUPET0	-0.682 (0.688)	0.3417	-0.050 (1.133)	0.9648
TIME*AGE	0.021 (0.011)	0.0591	0.023 (0.011)	0.0321
TIME*EDU 2	0.108 (0.125)	0.3887	0.110 (0.112)	0.329
TIME*EDU 3	0.242 (0.134)	0.0756	0.181 (0.122)	0.136
TIME*EDU 4	0.027 (0.172)	0.8776	-0.029 (0.149)	0.845
TIME*JOB 1	-0.471 (0.212)	0.0297	-0.439 (0.198)	0.0266
TIME*ADL	0.063 (0.029)	0.0345	0.057 (0.026)	0.0285
TIME*CDRSB0	-0.021 (0.004)	< .0001	-0.021 (0.004)	< .0001

3.1.2 Marginal Model

Now we want to study the impact of a MAR assumption on the marginal model for the dichotomized CDRSB quantity. In our previous analysis (see [7]) we investigated a Generalized Estimating Equations (GEE) model to our data, and we drew the conclusion that a quadratic interaction over time in the link function was suitable. Then, we also took into consideration the interaction of Time and Time² with the other available covariates, and we reduced the model based on the Quasi-Likelihood Information Criterion (QIC)

value, getting to a final model in the form

$$\begin{cases} Y_{ij} \sim \text{Bernoulli}(\pi_{ij}) \\ \text{Logit}(\pi_{ij}) = \log\left(\frac{\pi_{ij}}{1-\pi_{ij}}\right) = \eta_{ij} \\ \eta_{ij} = \beta_0 + \beta_{0,S}Sex_i + \beta_{0,B}BMI_i + \beta_{0,AD}ADL_{ij} \\ \quad + (\beta_1 + \beta_{1,B}BMI_i + \beta_{1,AD}ADL_{ij}) \cdot Year_{ij} \\ \quad + (\beta_2 + \beta_{2,AD}ADL_{ij}) \cdot Year_{ij}^2 \end{cases}$$

where $\pi_{ij} = \mathbb{P}(Y_{ij} = 1)$. Now we want to discuss how these results change when we introduce a MAR assumption.

First of all, as suggested in the book of Molenberghs [12], the estimates of a GEE model are correct only under an MCAR assumption on the missingness pattern of the data. To overcome this problem when we assume a MAR mechanism, two ways are proposed by the authors: Weighted GEE (WGEE) or MI followed by (multiple) GEE estimates, pooled together following the approach proposed by Rubin in 1978 [13] and refined by Rubin and Little in 2002 [10]. Beunckens et al. (2008, [3]) have proven that, even though WGEE is asymptotically unbiased, Generalized Estimating Equations with Multiple Imputation (MI-GEE) is less biased and more precise in small and moderate samples and MI-GEE is more robust to misspecification in either the imputation or measurement model. Following this very last property, we decide to apply a MI-GEE procedure, even if our available dataset has a consistent number of observations. Speaking about the number of imputations that we will perform, classic results adopted by Molenberghs and Verbeke [12] suggest that even a small number of imputations (5 or 10) is sufficient to get nice results. However, more recent results (Graham et al., 2007 [8]) suggest to use a number of 20 imputations in a dataset with almost 30% of missing data (as it is ours). In addition, we included all the covariates available in the dataset to perform the imputation task, following the results of Mainzer et al. (2024, [11]).

At this point, after choosing the imputation approach, we need to decide the form of the model we want to study. As we have done in our previous analysis ([7]), we model the interaction between the probabilities π_{ij} and the linear predictor $\eta_{ij} = \beta^T x_{ij}$ through the logit function

$$\log\left(\frac{\pi_{ij}}{1-\pi_{ij}}\right) = \eta_{ij}$$

Speaking about the order of the **Time** interaction in η_{ij} , we decide to consider again a quadratic model. As a matter of fact, even if the estimates previously obtained in [7] are not correct in a MAR context, they still provide a good starting point for the statistical analysis of our data and they show how the behavior of the temporal evolution would change over time considering different temporal orders. On the other hand, we cannot use the reduced interactions (in terms of time and other covariates) that we selected in our previous model, as the estimates of the latter are biased under a MAR assumption (they would be correct only under MNAR). So we start by considering all the interactions between **Time** and **Time**² with other covariates in the linear predictor η_{ij} in the MI-GEE model. As done in the previous analysis [7], we will use an unstructured working correlation matrix.

In Table 3 we see a comparison of the estimates (with their standard errors and p -values) from the MI-GEE and the GEE approaches with $M = 20$ imputations on the full-interaction model. A complete table with also the confidence intervals of the estimates of the MI case can be retrieved in Appendix A (see Table 7). Generally speaking, the estimates are quite similar in the two cases in absolute values, even if the standard errors computed under the MI case are higher. Said so, we also observe some differences between the MI-GEE and the GEE results. For example, some variables that under the GEE approach were significant (such as the interaction of **Time**² with **ADL**) become not-significant in the MI model, while some others (such as the interaction between time and the bio-markers) increase their significance in this second approach. In addition, the quadratic interaction over time seems to be less significant in the MI case, but it is still the parameter with the lowest p -value.

For the sake of completeness, in Table 8 in Appendix A we report the estimates of the reduced marginal model that we selected in the previous report [7] under the MI approach. We observe that the results were biased by the (implicit) MCAR assumption: as an example, the interaction of **ADL** with the squared temporal indication is now absolutely not-significant. This is another sign that our data do not follow an MCAR mechanism but a MAR (or MNAR) one.

Finally, it is interesting to see if we can reduce this model in the interactions present in η_{ij} . Since we cannot apply an Information Criterion approach (at least naively, as we are working with pooling estimates from different datasets), we simply perform a stepwise procedure in which we discard at each step those values whose pooled p -value is bigger than 0.1. We strictly adhere to the hierarchy principle: a main effect (Intercept) cannot be removed if its corresponding interaction with Time (Slope) remains significant. We get to the final model of the form

$$\begin{cases} Y_{ij} \sim \text{Bernoulli}(\pi_{ij}) \\ \log\left(\frac{\pi_{ij}}{1-\pi_{ij}}\right) = \eta_{ij} \\ \eta_{ij} = \beta_0 + \beta_{0,A}Age_i + \beta_{0,B}BMI_i + \beta_{0,T}TAU_i \\ \quad + (\beta_1 + \beta_{1,B}BMI_i) \cdot Year_{ij} + \beta_2 Year_{ij}^2 \end{cases}$$

Interestingly enough, the final model retains **Time**, **Time**², **Age**, **BMI**, **TAU** and the interaction of **BMI** with time. Comparing this result with the one we got in [7] we observe that using this model leads to keeping the Tau protein biomarker, which was not considered in the previous result. On the other hand, in the MI approach we loose significance of the **ADL** variable, which seemed to be an interesting quantity to study. In addition, with the MI-GEE procedure we get to a final model that resembles more closely the Generalized Linear Mixed-Effects Model (GLMM) approach (except for the presence of the quadratic interaction of the temporal variable).

Table 3: Comparison of the results for the full quadratic MI-GEE model (with $M = 20$ imputations) and the naive GEE model.

Parameter	MI-GEE ($M = 20$)		GEE	
	Estimate (SE)	p -value	Estimate (SE)	p -value
Intercept	-0.997 (0.241)	< .001	-0.948 (0.197)	< .001
TIME	0.139 (0.133)	0.294	-0.051 (0.112)	0.651
TIME*TIME	0.034 (0.018)	0.066	0.070 (0.016)	< .001
SEX	0.175 (0.129)	0.173	0.184 (0.128)	0.151
AGE_STD	0.119 (0.144)	0.413	0.024 (0.112)	0.829
BMI_STD	0.101 (0.062)	0.101	0.104 (0.061)	0.088
ADL	0.010 (0.028)	0.710	0.008 (0.028)	0.762
TAUPET_STD	-0.093 (0.461)	0.842	-0.053 (0.087)	0.539
ABPET_STD	-0.134 (0.141)	0.348	0.006 (0.075)	0.932
TIME*SEX	-0.044 (0.077)	0.570	-0.067 (0.073)	0.359
TIME*AGE_STD	-0.180 (0.119)	0.139	-0.089 (0.068)	0.188
TIME*BMI_STD	-0.044 (0.034)	0.195	-0.034 (0.035)	0.331
TIME*ADL	-0.006 (0.018)	0.754	0.012 (0.015)	0.445
TIME*TAUPET_STD	0.209 (0.203)	0.313	0.056 (0.051)	0.270
TIME*ABPET_STD	0.190 (0.150)	0.219	0.041 (0.056)	0.466
TIME*TIME*SEX	-0.001 (0.012)	0.944	0.004 (0.011)	0.687
TIME*TIME*AGE_STD	0.024 (0.024)	0.322	0.016 (0.010)	0.108
TIME*TIME*BMI_STD	0.002 (0.005)	0.681	-0.001 (0.005)	0.842
TIME*TIME*ADL	-0.000 (0.003)	0.920	-0.004 (0.002)	0.085
TIME*TIME*TAUPET_STD	-0.033 (0.024)	0.170	-0.007 (0.007)	0.289
TIME*TIME*ABPET_STD	-0.028 (0.026)	0.284	-0.009 (0.009)	0.301

3.1.3 Random-Effects Model

A Generalized Linear Mixed-effects Model was formulated to analyze the longitudinal trajectories of CDRSB_CAT (where $\pi_{ij} = \mathbb{P}(Y_{ij} = 1)$ denotes the probability of the event), ensuring consistent and unbiased parameter estimates under the MAR assumption for the outcome variable. Following our methodological framework, we specified a structure with both a random intercept and a random slope to capture inherent inter-individual variability. Although model reduction techniques are often employed to simplify interpretation, the full model (including all primary predictors and their interactions with time) was retained in this study. This approach ensures that all theoretically relevant factors and potential confounders are accounted for, providing a comprehensive estimation of the fixed and random effects:

Table 4: Results for the MI-GEE model with $M = 20$ imputations reduced using only variables whose pooled p -value is smaller than 0.1.

Parameter	Estimate (SE)	95% CI	p -value
Intercept	-0.775 (0.062)	-0.897 – -0.653	< .0001
TIME	0.060 (0.038)	-0.014 – 0.135	0.1125
TIME*TIME	0.031 (0.006)	0.018 – 0.044	< .0001
AGE_STD	-0.069 (0.039)	-0.149 – 0.011	0.0884
BMI_STD	0.103 (0.060)	-0.015 – 0.222	0.0870
TAUPET_STD	0.105 (0.061)	-0.019 – 0.229	0.0938
TIME*BMI_STD	-0.033 (0.020)	-0.071 – 0.006	0.0999

$$\left\{ \begin{array}{l} Y_{ij} \sim \text{Bernoulli}(\pi_{ij}) \\ \text{Logit}(\pi_{ij}) = \log\left(\frac{\pi_{ij}}{1-\pi_{ij}}\right) = \eta_{ij} \\ \eta_{ij} = \beta_0 + \beta_{0,A}\text{Age}_i + \beta_{0,B}\text{BMI}_i + \beta_{0,S}\text{Sex}_i \\ \quad + \beta_{0,AD}\text{ADL}_i + \beta_{0,AB}\text{ABPET}_{ij} + \beta_{0,T}\text{TAU}_{ij} \\ \quad + \left(\beta_1 + \beta_{1,A}\text{Age}_i + \beta_{1,B}\text{BMI}_i + \beta_{1,S}\text{Sex}_i \right. \\ \quad \left. + \beta_{1,AD}\text{ADL}_i + \beta_{1,AB}\text{ABPET}_{ij} + \beta_{1,T}\text{TAU}_{ij}\right) \cdot \text{Year}_{ij} \\ \quad + b_{0i} + b_{1i} \cdot \text{Year}_{ij} \\ \left(\begin{matrix} b_{0i} \\ b_{1i} \end{matrix}\right) \sim N\left(\left(\begin{matrix} 0 \\ 0 \end{matrix}\right), \left(\begin{matrix} d_{11} & d_{12} \\ d_{21} & d_{22} \end{matrix}\right)\right) \end{array} \right.$$

While standard GLMMs are valid under the MAR assumption for the outcome variable, a critical limitation arises when missingness extends to the covariates. Ideally, one might consider applying the GLMM directly to the observed data; however, standard software implementations typically default to a complete-case analysis when predictors are missing. As noted by Huque et al. in [9], a common misconception is that mixed models automatically handle all MAR data. In reality, if covariates contain missing values, excluding these subjects leads to potential selection bias and a significant loss of statistical efficiency.

To address this, we implemented a Generalized Linear Mixed-Effects Model with Multiple Imputation (MI-GLMM) prior to the longitudinal analysis. Following the recommendations of Huque [9], we utilized MI to impute missing values in both covariates and outcomes. This strategy allows us to retain the full sample size and maximize information, ensuring that the subsequent mixed-effects analysis is both robust and unbiased. Regarding the specification of the imputation process, the same conclusions derived for the previous models hold true. Even if an higher number of imputations is suggested following results from Graham et al. (2007, [8]), we selected a total of 10 imputations

($M = 10$) to reduce convergence problems.

The quantitative impact of this methodological choice is presented in Table 5, which contrasts the results of the complete-case analysis against the pooled estimates from the MI.

Table 5: Comparison: Estimates (SE) and p-values: Original GLMM vs MI-GLMM ($M = 10$).

Parameter	GLMM		MI-GLMM ($M = 10$)	
	Estimate (SE)	p-value	Estimate (SE)	p-value
Intercept	-3.3429 (0.5131)	<.0001	-1.5788 (0.3467)	<.0001
AGE_STD	0.4139 (0.2757)	0.1334	0.2477 (0.1926)	0.2012
BMI_STD	0.2790 (0.1578)	0.0772	0.1216 (0.0953)	0.2022
SEX (0)	0.4956 (0.3283)	0.1313	-0.2444 (0.2006)	0.2231
ADL	0.0541 (0.0698)	0.4386	0.0243 (0.0434)	0.5758
ABPET_STD	-0.1035 (0.1763)	0.5570	-0.1311 (0.1397)	0.3540
TAUPET_STD	-0.2223 (0.1269)	0.0799	0.1034 (0.1536)	0.5140
TIME	0.8113 (0.1595)	<.0001	0.4316 (0.1063)	<.0001
TIME · BMI_STD	-0.0844 (0.0496)	0.0885	-0.0395 (0.0298)	0.1854
TIME · TAUPET_STD	0.0776 (0.0315)	0.0139	-0.0195 (0.0355)	0.5926
TIME · SEX (0)	-0.1356 (0.1030)	0.1880	0.0614 (0.0613)	0.3161
TIME · AGE_STD	-0.1631 (0.0855)	0.0564	-0.0922 (0.0671)	0.1773
TIME · ABPET_STD	0.0116 (0.0530)	0.8264	0.0374 (0.0515)	0.4771
TIME · ADL	-0.0178 (0.0216)	0.4084	-0.0087 (0.0132)	0.5099

The most striking difference observed is the significant shift in parameter estimates, particularly for the **TIME** variable, which decreases from 0.8113 to 0.4316, and the **Intercept**, moving from -3.3429 to -1.5788 . Under the MAR assumption, this reduction in the **TIME** coefficient suggests that the original complete-case model was suffering from selection bias. By excluding participants with incomplete covariate data, the original model overestimated the rate of clinical progression. The MI approach corrects this by incorporating a more representative sample, resulting in a smoother and more generalizable trajectory of the population.

Furthermore, this comparison demonstrates a clear gain in statistical power. While both models maintain high significance for primary effects ($p < .0001$), the MI-GLMM shows a substantial reduction in standard errors (SE). For instance, the SE for the **Intercept** drops from 0.5131 to 0.3467, and the SEs for other covariates are nearly halved. This gain in precision confirms that MI effectively recovered valuable information from partially observed cases that the standard mixed model discarded. These results empirically validate the theoretical recommendations of Huque et al., confirming that addressing missing covariates is essential to ensure that fixed effect estimates are both unbiased and efficient.

Having established the efficiency and unbiasedness of the parameter estimates under

the MI-GLMM framework, we proceeded to validate the model’s structural assumptions by examining the variance trends.

In a binomial GLMM, the variance is not a constant parameter (homoscedasticity) but is functionally dependent on the mean, defined by the relationship $\text{Var}(Y) = \hat{\pi}(1 - \hat{\pi})$, where $\hat{\pi}$ is the predicted probability. As illustrated in Figure 8, both the fitted and empirical variance curves exhibit the characteristic inverted-U shape expected in this distribution. This pattern reflects that uncertainty is maximal when the predicted probability is near 0.5 (around Year 4) and decreases as subjects approach the boundaries of the outcome (0 or 1).

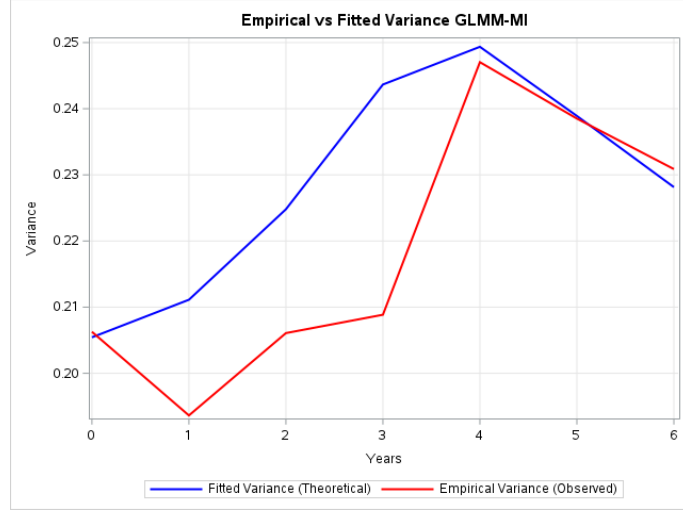


Figure 8: Dichotomized CDRSB’s Empirical Variance vs. Fitted Variance (MI-GLMM).

The visual comparison in Figure 8 serves as a robust diagnostic tool. The convergence of the two lines in the latter half of the study indicates that the model is well-calibrated and correctly accounts for the population’s heterogeneity. However, a structural divergence is evident during the initial years. This gap arises because the GLMM relies on a linear time trend within the logit link function. While this transformation produces a non-linear probability curve (sigmoid), the underlying linear predictor mathematically assumes a continuous, monotonic accumulation of risk from baseline. Consequently, the model cannot fully reproduce the lag phase of stability observed empirically, where subjects exhibit a delayed onset of progression. Although this implies that the model slightly overestimates early variability, the strong alignment in subsequent years confirms that this approximation is a necessary trade-off for parsimony, successfully capturing the global longitudinal trend without overfitting the initial noise.

We compared the empirical mean trajectory (Figure 9a) with the pooled prediction derived from the MI-GLMM (Figure 9b). Both curves exhibit a consistent progression from approximately 30% prevalence at baseline to nearly 65% at Year 6. However, while the empirical data displays a distinct plateau or lag phase during the first three years, the MI-GLMM approximates this with a smooth, monotonic increase due to the underlying linear time effect.

Crucially, the shaded region in Figure 9b represents the 95% Confidence Interval (CI) of the predicted mean, distinct from the observed variance bands in Figure 9a. The bow-

tie shape of the CI, which narrows in the center (Years 2-4) and widens at the extremes, reflects the inherent property of regression models where estimation uncertainty increases as we move away from the temporal center of the data. Despite smoothing the initial lag, the model successfully captures the global rate of decline and aligns with the observed endpoints.

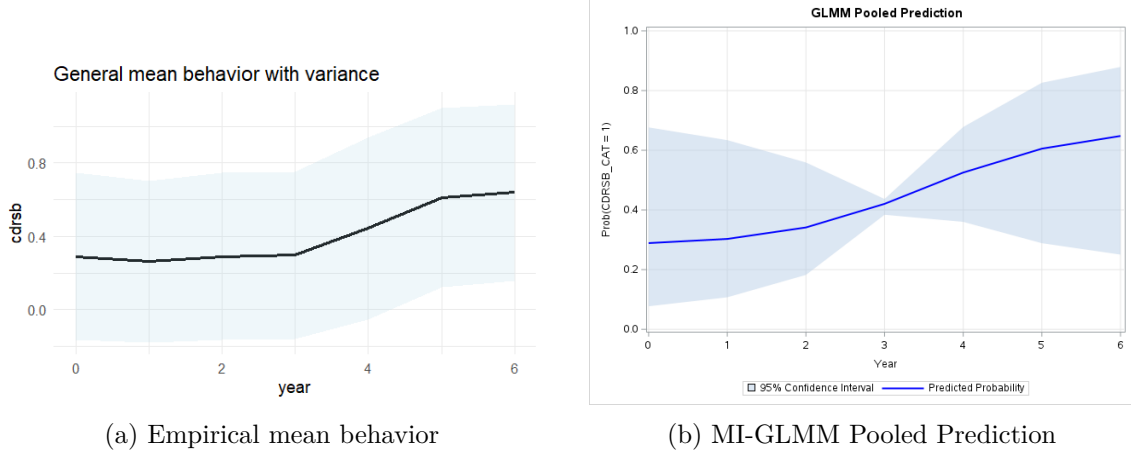


Figure 9: Comparison of the observed probability trajectory vs. the fitted model prediction.

Finally, we reconciled the visual patterns observed in the exploratory phase with the multivariable adjustments provided by the model. While the raw data initially suggested a potential divergence in progression rates based on baseline ABPET levels, the MI-GLMM revealed a non-significant interaction term for $\text{TIME} \cdot \text{ABPET_STD}$ ($p = 0.8264$). This discrepancy implies that the observed visual differences were likely confounded by other covariates, such as **Age** or **BMI**, which the multivariable model successfully deconvoluted. Furthermore, the robust positive estimate for **TIME** (0.43) aligns with the overall increasing trend in disease severity observed in the aggregate plots. This confirms that, despite the minor structural mismatch during the initial variance lag, the linear approximation correctly captures the global directionality and magnitude of the disease progression.

Collectively, these diagnostic evaluations validate the MI-GLMM as a robust analytical framework for this study. The quantitative comparison with the complete-case analysis demonstrated that MI effectively mitigated selection bias and improved statistical efficiency, as evidenced by the significant correction of the **TIME** coefficient and the substantial reduction in standard errors. Structurally, the convergence of empirical and fitted variances confirms that the binomial assumptions hold, while the trajectory analysis shows that the model accurately tracks the long-term clinical decline. Although the assumption of linearity on the logit scale introduces a slight overestimation of risk during the initial lag phase, this approximation is theoretically justified and does not compromise the validity of the global parameter estimates. Consequently, the results derived from this model can be interpreted with confidence as reliable reflections of the underlying biological associations, minimizing the distortions typically caused by missing data.

3.2 Sensitivity Analysis regarding MNAR in the LME model

In the last part of our document we would like to perform a (brief) Sensitivity Analysis (SA) to assess the potential impact of MNAR on the BPRS quantity. In order to do that, we will make use of the `mice` package available in R.

Specifically, we employ a Delta-adjustment method (see [4, 12] and [14] for theoretical background), which adds a fixed parameter δ to the imputed BPRS scores. This simulates a scenario where patients who drop out experience a steeper decline than predicted by the observed data. We compare two imputation strategies. The first applies the shift δ uniformly to all imputed values, assuming a shared underlying mechanism. The second adopts a Pattern-Mixture Model (PMM) framework using the Neighbouring Case Missing Values (NCMV) restriction. Under NCMV, missing data for patients dropping out at time t are imputed using information from patients dropping out at time $t + 1$. While other restrictions such as Complete Case Missing Values (CCMV) or Available Case Missing Values (ACMV) exist, we selected NCMV because it provides a conservative sensitivity analysis that departs from the standard MAR assumption, unlike ACMV which typically aligns with MAR.

We examine the shift parameter δ over a range from 0 (corresponding to a MAR assumption) to 6 (reflecting severe MNAR), in increments of 2 (smaller increments were taken into consideration, here we report an higher value to summarize the obtained results). It is important to note that, consistent with the scale definition [1], we cap the imputed BPRS scores at a maximum of 168. Furthermore, we limited the shift to $\delta = 6$ because higher values yielded biologically implausible estimates for early dropouts. To compare the results, we fit the reduced linear mixed-effects model obtained in Section 3.1.1 for each scenario to observe if the main conclusions remained stable. Usually this strategy would include studying the threshold value of δ that makes a treatment become not significant: however, since in our case we do not have explicitly a reference treatment (or reference biological quantities, since the biomarkers' levels are not included in our reduced model), we will focus on the value of the coefficient linked to the (linear) time. A summary of the estimates (and their standard errors) can be seen in Table 6, while a complete report is given in Table 9 in Appendix A.

The linear time coefficient (β_{year}) shows distinct behaviors between the two strategies. Under the Standard Delta-Adjustment (STD), the estimate remains relatively stable, increasing moderately from 5.21 (Shift 0) to 6.21 (Shift 6). This suggests that if the missing data mechanism is uniform across all patients, the impact of unobserved worsening is limited. Conversely, the PMM reveals a high sensitivity to the MNAR assumption. As the shift parameter δ increases, the estimated yearly progression rises drastically, reaching 10.59 at $\delta = 6$. This indicates that if we assume early dropouts follow a fundamentally different (and worse) trajectory than completers, the "true" disease progression could be significantly faster than what is observed under MAR assumptions.

A notable "tipping point" is observed in the interaction term between the ADL variable and time. In the standard model, this coefficient remains positive across all shifts, implying that patients with higher baseline functional impairment (higher ADL) consistently experience a slightly faster cognitive decline. However, in the PMM, this relationship inverts as the shift increases, becoming negative. This reversal suggests that under severe MNAR conditions combined with structural dropout patterns, the relationship between baseline function and subsequent decline may be distorted or fundamentally different.

Table 6: Sensitivity Analysis: Model Estimates of some of the most relevant parameters under Standard Delta-Adjustment vs. Pattern-Mixture Models.

Term	Standard Delta-Adjustment (STD)				Pattern-Mixture Model (PMM)			
	Shift 0	Shift 2	Shift 4	Shift 6	Shift 0	Shift 2	Shift 4	Shift 6
(Intercept)	75.788 (sd 1.11) p< .001	75.991 (sd 1.17) p< .001	76.140 (sd 1.27) p< .001	76.213 (sd 1.36) p< .001	75.634 (sd 1.19) p< .001	74.883 (sd 1.24) p< .001	74.177 (sd 1.33) p< .001	73.260 (sd 1.54) p< .001
year	5.209 (sd 0.47) p< .001	5.544 (sd 0.54) p< .001	5.863 (sd 0.45) p< .001	6.212 (sd 0.47) p< .001	5.252 (sd 0.42) p< .001	7.194 (sd 0.41) p< .001	9.013 (sd 0.41) p< .001	10.598 (sd 0.40) p< .001
adl_num	-0.092 (sd 0.15) p=.525	-0.145 (sd 0.15) p=.337	-0.193 (sd 0.16) p=.236	-0.233 (sd 0.17) p=.183	-0.066 (sd 0.15) p=.662	-0.000 (sd 0.16) p=.999	0.045 (sd 0.17) p=.791	0.126 (sd 0.20) p=.524
adl_num:yr	0.207 (sd 0.05) p< .001	0.186 (sd 0.05) p< .001	0.172 (sd 0.05) p=.001	0.152 (sd 0.05) p=.006	0.196 (sd 0.04) p< .001	0.004 (sd 0.04) p=.935	-0.193 (sd 0.05) p< .001	-0.359 (sd 0.05) p< .001
cdrsb_base	-0.019 (sd 0.01) p=.011	-0.013 (sd 0.01) p=.106	-0.006 (sd 0.01) p=.472	0.000 (sd 0.01) p=.976	-0.021 (sd 0.01) p=.005	-0.018 (sd 0.01) p=.016	-0.013 (sd 0.01) p=.136	-0.007 (sd 0.01) p=.477
cdrsb_b:yr	0.021 (sd 0.01) p< .001	-0.023 (sd 0.01) p< .001	-0.026 (sd 0.01) p< .001	-0.029 (sd 0.01) p< .001	0.020 (sd 0.01) p< .001	-0.016 (sd 0.01) p=.001	-0.013 (sd 0.01) p=.022	-0.011 (sd 0.01) p=.091

4 Conclusion and further studies

The analysis performed in this document highlights that the use of MI within the LME model does not produce results that differ significantly from those obtained with the model estimated without imputation. This suggests that a MAR assumption may be adequate to describe the missingness mechanism in the data, and that the standard LME model alone is sufficient to draw reliable conclusions. In contrast, when MI is combined with the GLMM, slight differences are observed in the estimated coefficients and standard errors, indicating that the MAR assumption may not be fully satisfied in this context. Regarding the GEE model, since it cannot directly handle MAR or MNAR mechanisms, MI was preferred over the WGEE approach, which is considered less robust in this application. Accordingly, the observed differences in coefficients and standard errors are meaningful, reflecting the inability of the GEE model to properly account for the missingness pattern present in the data. Finally, a sensitivity analysis was conducted to assess the potential presence of an MNAR mechanism. The results show only minimal changes in coefficients and p-values, suggesting that the MNAR assumption cannot be fully confirmed.

From a clinical perspective, the analyses across the different modeling frameworks (LME, GEE, and GLMM) lead to consistent conclusions on the determinants of disease progression. Across models, changes in both cognitive (CDRSB) and psychiatric (BPRS) outcomes were primarily associated with baseline demographic and functional characteristics. While exploratory analyses suggested a potential association with biomarkers such as Tau and Abpet, these effects were markedly reduced after adjustment for other covariates. Overall, the results indicate that, in this cohort, clinical and functional measures pro-

vide more stable prognostic information than the available biomarker profiles, supporting the relevance of functional assessment in the longitudinal evaluation of Alzheimer’s disease.

For future studies, combining MI with validation techniques such as bootstrap or cross-validation could help strengthen the reliability of the results. Bootstrapping helps clarify the variability of parameter estimates, while cross-validation evaluates how stable and generalizable the model is across different data partitions. Together, these techniques provide a more thorough assessment of the robustness of conclusions when missing data are involved.

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A Appendix

Table 7: Results for the MI-GEE model with $M = 20$ imputations.

Parameter	Estimate (SE)	95% CI	<i>p</i> -value
Intercept	-0.997 (0.241)	[-1.471, -0.523]	< .0001
TIME	0.139 (0.133)	[-0.122, 0.400]	0.2943
TIME*TIME	0.034 (0.018)	[-0.002, 0.070]	0.0663
SEX	0.175 (0.129)	[-0.077, 0.428]	0.1734
AGE_STD	0.119 (0.144)	[-0.169, 0.407]	0.4132
BMI_STD	0.101 (0.062)	[-0.020, 0.222]	0.1006
ADL	0.010 (0.028)	[-0.044, 0.064]	0.7095
TAUPET_STD	-0.093 (0.461)	[-1.038, 0.853]	0.8423
ABPET_STD	-0.134 (0.141)	[-0.418, 0.151]	0.3481
TIME*SEX	-0.044 (0.077)	[-0.196, 0.108]	0.5698
TIME*AGE_STD	-0.180 (0.119)	[-0.422, 0.062]	0.1387
TIME*BMI_STD	-0.044 (0.034)	[-0.110, 0.023]	0.1951
TIME*ADL	-0.006 (0.018)	[-0.041, 0.029]	0.7536
TIME*TAUPET_STD	0.209 (0.203)	[-0.206, 0.624]	0.3132
TIME*ABPET_STD	0.190 (0.150)	[-0.120, 0.500]	0.2190
TIME*TIME*SEX	-0.001 (0.012)	[-0.025, 0.023]	0.9440
TIME*TIME*AGE_STD	0.024 (0.024)	[-0.025, 0.073]	0.3218
TIME*TIME*BMI_STD	0.002 (0.005)	[-0.008, 0.012]	0.6809
TIME*TIME*ADL	-0.000 (0.003)	[-0.006, 0.005]	0.9199
TIME*TIME*TAUPET_STD	-0.033 (0.024)	[-0.081, 0.015]	0.1703
TIME*TIME*ABPET_STD	-0.028 (0.026)	[-0.081, 0.025]	0.2843

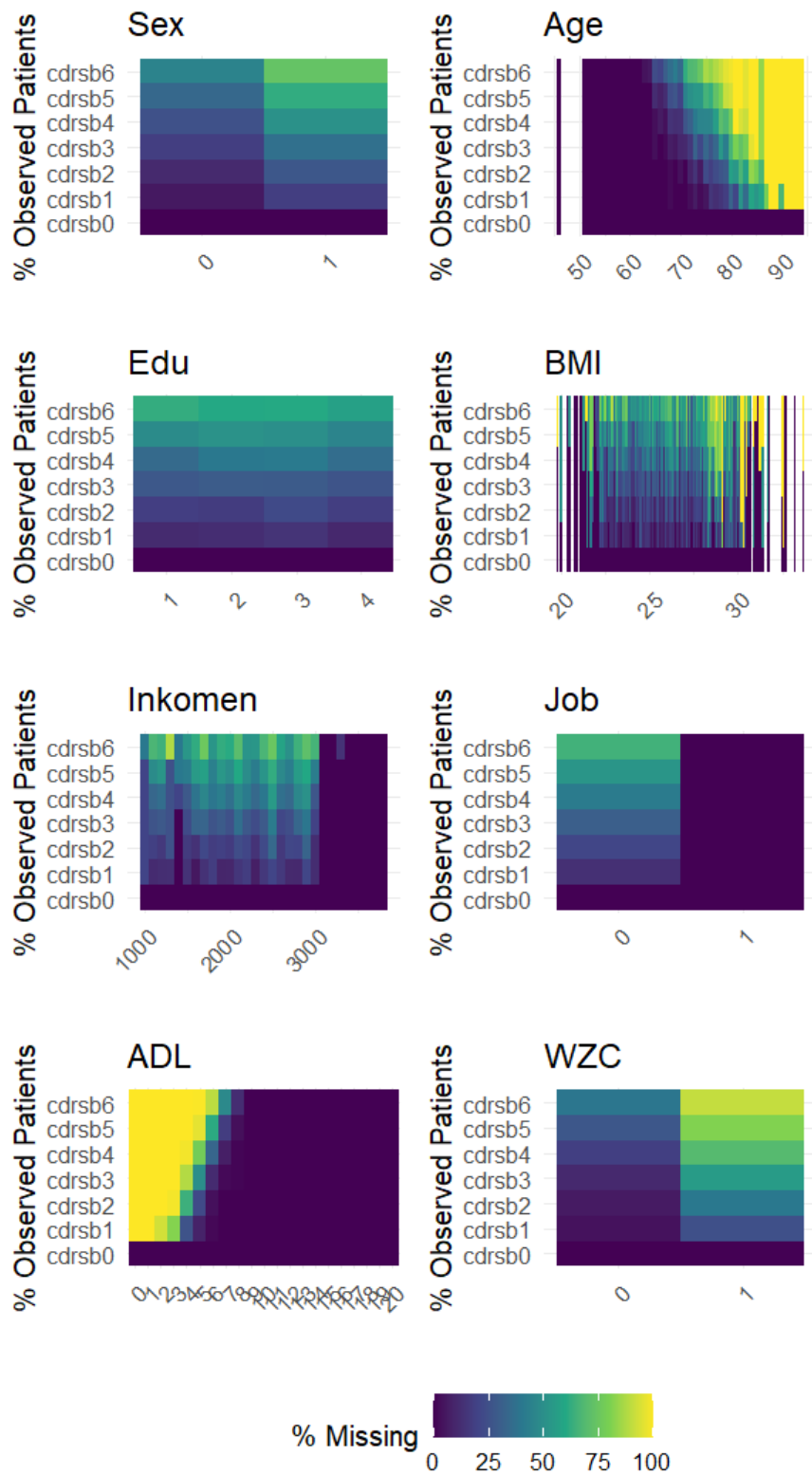


Figure 10: Distribution of Missingness by Baseline Covariates. Darker colors represent observed data, while lighter colors (yellow) indicate missing values/dropout.

Table 8: Results for the MI-GEE model with $M = 20$ imputations on the reduced marginal model that we obtained in Report 2.

Parameter	Estimate (SE)	95% CI	<i>p</i> -value
Intercept	-0.870 (0.147)	-1.158 – -0.582	< .0001
TIME	0.016 (0.114)	-0.210 – 0.243	0.8867
TIME*TIME	0.045 (0.023)	-0.001 – 0.091	0.0550
SEX	0.032 (0.042)	-0.050 – 0.115	0.4416
BMI_STD	0.103 (0.061)	-0.016 – 0.222	0.0907
ADL	0.009 (0.019)	-0.029 – 0.046	0.6527
TIME*BMI_STD	-0.032 (0.020)	-0.071 – 0.007	0.1103
TIME*ADL	0.005 (0.014)	-0.023 – 0.032	0.7335
TIME*TIME*ADL	-0.001 (0.003)	-0.007 – 0.004	0.5797

Table 9: Sensitivity Analysis: Full Model Estimates (Standard vs Pattern-Mixture Models).

Term	Standard Delta-Adjustment (STD)				Pattern-Mixture Model (PMM)			
	Shift 0	Shift 2	Shift 4	Shift 6	Shift 0	Shift 2	Shift 4	Shift 6
(Intercept)	75.788 (sd 1.11) p< .001	75.991 (sd 1.17) p< .001	76.140 (sd 1.27) p< .001	76.213 (sd 1.36) p< .001	75.634 (sd 1.19) p< .001	74.883 (sd 1.24) p< .001	74.177 (sd 1.33) p< .001	73.260 (sd 1.54) p< .001
age_std	12.549 (sd 0.39) p< .001	12.571 (sd 0.40) p< .001	12.585 (sd 0.42) p< .001	12.614 (sd 0.45) p< .001	12.562 (sd 0.37) p< .001	12.640 (sd 0.38) p< .001	12.732 (sd 0.43) p< .001	12.991 (sd 0.49) p< .001
edu2	0.062 (sd 0.20) p=.760	0.047 (sd 0.21) p=.826	0.058 (sd 0.24) p=.805	0.048 (sd 0.25) p=.851	0.076 (sd 0.18) p=.675	0.058 (sd 0.19) p=.763	0.080 (sd 0.22) p=.713	0.061 (sd 0.26) p=.814
edu3	-0.078 (sd 0.18) p=.674	-0.037 (sd 0.20) p=.855	0.025 (sd 0.22) p=.911	0.073 (sd 0.25) p=.762	-0.050 (sd 0.17) p=.768	-0.054 (sd 0.18) p=.763	-0.004 (sd 0.20) p=.985	0.038 (sd 0.24) p=.875
edu4	0.141 (sd 0.18) p=.433	0.159 (sd 0.19) p=.405	0.198 (sd 0.21) p=.349	0.216 (sd 0.24) p=.363	0.132 (sd 0.17) p=.442	0.119 (sd 0.18) p=.516	0.147 (sd 0.20) p=.474	0.165 (sd 0.24) p=.494
bmi_std	0.453 (sd 0.08) p< .001	0.448 (sd 0.08) p< .001	0.442 (sd 0.08) p< .001	0.441 (sd 0.09) p< .001	0.457 (sd 0.07) p< .001	0.444 (sd 0.08) p< .001	0.447 (sd 0.09) p< .001	0.458 (sd 0.17) p< .001
job1	-3.996 (sd 0.99) p< .001	-3.433 (sd 1.03) p=.001	-2.898 (sd 1.11) p=.009	-2.408 (sd 1.20) p=.045	-4.228 (sd 1.02) p< .001	-4.399 (sd 1.06) p< .001	-4.399 (sd 1.17) p< .001	-4.609 (sd 1.34) p< .001
adl_num	-0.092 (sd 0.15) p=.525	-0.145 (sd 0.15) p=.337	-0.193 (sd 0.16) p=.236	-0.233 (sd 0.17) p=.183	-0.066 (sd 0.15) p=.662	-0.000 (sd 0.16) p=.999	0.045 (sd 0.17) p=.791	0.126 (sd 0.20) p=.524
wzc1	1.995 (sd 0.15) p< .001	2.065 (sd 0.15) p< .001	2.154 (sd 0.14) p< .001	2.264 (sd 0.14) p< .001	2.065 (sd 0.13) p< .001	2.025 (sd 0.13) p< .001	2.131 (sd 0.14) p< .001	2.352 (sd 0.17) p< .001
cdrsb_b	-0.019 (sd 0.01) p=.011	-0.013 (sd 0.01) p=.106	-0.006 (sd 0.01) p=.472	0.000 (sd 0.01) p=.976	-0.021 (sd 0.01) p=.005	-0.018 (sd 0.01) p=.016	-0.013 (sd 0.01) p=.136	-0.007 (sd 0.01) p=.477
year	5.209 (sd 0.47) p< .001	5.544 (sd 0.54) p< .001	5.863 (sd 0.45) p< .001	6.212 (sd 0.47) p< .001	5.252 (sd 0.42) p< .001	7.194 (sd 0.41) p< .001	9.013 (sd 0.41) p< .001	10.598 (sd 0.40) p< .001
age_std:yr	-0.097 (sd 0.13) p=.478	-0.088 (sd 0.14) p=.538	-0.075 (sd 0.15) p=.615	-0.078 (sd 0.16) p=.628	-0.170 (sd 0.08) p=.037	-0.173 (sd 0.07) p=.023	-0.242 (sd 0.07) p< .001	-0.347 (sd 0.08) p< .001
edu2:yr	0.191 (sd 0.15) p=.193	0.207 (sd 0.15) p=.166	0.200 (sd 0.16) p=.208	0.208 (sd 0.16) p=.208	0.204 (sd 0.13) p=.117	0.190 (sd 0.12) p=.128	0.163 (sd 0.13) p=.215	0.174 (sd 0.15) p=.238
edu3:yr	0.276 (sd 0.13) p=.036	0.276 (sd 0.14) p=.045	0.249 (sd 0.14) p=.087	0.232 (sd 0.15) p=.130	0.240 (sd 0.11) p=.038	0.267 (sd 0.11) p=.019	0.301 (sd 0.12) p=.011	0.327 (sd 0.14) p=.016
edu4:yr	0.079 (sd 0.13) p=.576	0.079 (sd 0.13) p=.556	0.061 (sd 0.14) p=.666	0.052 (sd 0.15) p=.733	0.050 (sd 0.12) p=.667	0.052 (sd 0.12) p=.650	0.061 (sd 0.12) p=.599	0.084 (sd 0.14) p=.543
job1:yr	-1.506 (sd 0.36) p< .001	-1.510 (sd 0.36) p< .001	-1.553 (sd 0.36) p< .001	-1.562 (sd 0.38) p< .001	-1.424 (sd 0.30) p< .001	-0.303 (sd 0.30) p=.313	0.737 (sd 0.32) p=.023	1.623 (sd 0.32) p< .001
adl_n:yr	0.207 (sd 0.05) p< .001	0.186 (sd 0.05) p< .001	0.172 (sd 0.05) p=.001	0.152 (sd 0.05) p=.006	0.196 (sd 0.04) p< .001	0.004 (sd 0.04) p=.935	-0.193 (sd 0.05) p< .001	-0.359 (sd 0.05) p< .001
cdrsb:yr	0.021 (sd 0.01) p< .001	-0.023 (sd 0.01) p< .001	-0.026 (sd 0.01) p< .001	-0.029 (sd 0.01) p< .001	0.020 (sd 0.01) p< .001	-0.016 (sd 0.01) p=.001	-0.013 (sd 0.01) p=.022	-0.011 (sd 0.01) p=.091